

Closed Graph Theorem:

If $T: \underline{X} \rightarrow \underline{Y}$ is linear op. b/w banach spaces.

T.F.A.E.:

- ① T is cont.
- ② Graph of T is closed in $X \times Y$ Product top.
" $\{ (x, Tx) : x \in X \}$

$$T \parallel \begin{matrix} X \\ \downarrow \\ I: (\ell^p, \|\cdot\|) \end{matrix} \longrightarrow \begin{matrix} Y \\ \downarrow \\ (\ell^p, \|\cdot\|_p) \end{matrix}$$

$$\{ (f, Tf) \}$$

$$\{ \underline{(f, f)} \}$$

$$(f_n, Tf_n) \longrightarrow (f, g)$$

$$\Rightarrow f_n \xrightarrow{\|\cdot\|} f \quad \overset{①}{\Rightarrow} \{ (f, Tf) \}$$

$$\Rightarrow Tf_n \xrightarrow{\|\cdot\|_p} g \Rightarrow \lim_{n \rightarrow \infty} f_n(k) = f(k)$$

$$\Rightarrow \underline{f_n} \xrightarrow{\|\cdot\|_p} g$$

$$\Rightarrow \lim_{n \rightarrow \infty} f_n(k) = g(k) = f(k)$$

$$\Rightarrow \underline{(g = f)}$$

$\Rightarrow$ graph is closed

$\Rightarrow$ I is bounded.

$\Rightarrow$ By bounded inverse thm,
 $\|\cdot\|$ & $\|\cdot\|_p$ are equivalent.

